Student's Name: .....

Arusha Regional Annual Examination December 2022

PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
ARUSHA REGION
FORM ONE ANNUAL EXAMINATION
CHEMISTRY

Time: 2:30 Hours **29th NOVEMBER 2022**

Instructions:

1. This paper consists of sections A and B with a total of ten (10) questions.
2. Answer ALL questions in the spaces provided.
3. All writing must be in blue or black ink, except drawing, which must be in pencil.
4. Cellular phones and any unauthorized materials are **not** allowed in the examination room.
5. Write your **NAME** at the top of every page.

SECTION A (15 Marks) Answer ALL Questions

1. Choose the most correct answer among the given alternatives and write down the answer in the space given.

(i) Apart from the Bunsen burner, which among the following heat sources is most suitable to use in the chemistry laboratory? A. Fire wood B. Kerosene stove C. Charcoal stove D. Candle

(ii) Solvent can be obtained from a solution by

A. Evaporation followed by decantation
B. Filtration and condensation
C. Evaporation and filtration
D. Evaporation and condensation

(iii)Which of the following processes occurs when wet clothe are drying?

A. Sublimation B. Filtration C. Condensation D. Evaporation

(iv) What is the use of pain killers found in the first aid kit?

A. Killing harmful germs B. Act as an antibiotic C. Relieve pain D. Cure malaria

(v) Fractional distillation process of a mixture of water and ethanol is possible because

A. Water and ethanol have the same boiling point. B. Water has lower boiling point than ethanol.
C. Ethanol has lower boiling point than water. D. Water and ethanol form partially immiscible liquid solution.

(vi) What is the use of wire gauze in the chemistry laboratory?

A. Covering bad wounds resulting from accidents in the laboratory
B. Covering beakers to prevent contamination of chemicals by flies
C. Allowing gas to pass through from one container to another
D. Distributing heat evenly during heating

(vii) One of the following gases is not a constituent of air:

A. Hydrogen B. Nitrogen C. Oxygen D. Argon

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(viii) Which conditions are necessary for an iron nail to rust?

A. Oxygen and moisture B. Nitrogen and heat C. Heat and oxygen D. Moisture and carbon dioxide

(ix) In a scientific procedure, the stage of accepting or rejecting the hypothesis is known as:

A. Problem identification B. Intelligent guess C. Experiment D. Conclusion

(x) The following is not a set of products made from the industrial application of chemistry.

A. Tooth paste and soap B. Rock stone and rain water. C. Soap and fertilizers D. Fertilizers and antiseptics

ANSWERS

i	ii	iii	iv	v	vi	vii	viii	ix	x

2. Match items in list A with the correct phrase in list B.

LIST A	**LIST B**
(i) A direct change from solid to gaseous state. (ii) A change of state from liquid to solid. (iii) The direct change from gaseous to solid state. (iv) A change in state of a substance from gas (vapour) to liquid. (v) A change in state of a substance from solid to liquid.	A. Melting B. Freezing C. Deposition D. Hypothesis formulation E. Condensation F. Experiment G. Sublimation H. Conclusi

ANSWERS

List A	i	ii	iii	iv	v
List B					

SECTION B (70 Marks) Answer all questions

3. (a) Draw a well labeled diagram of the Bunsen burner.

(b) State five differences between non-luminous and luminous flame

	Non-luminous flame	Luminous
1		
2		
3		
4		
5		

4. Draw a warning sign diagram that would appear on the storage containers of the following substances:
(i) Mercury

(ii) Concentrated sulphuric acid

(iii) Petrol

(iv) Sodium metal

(v) Hydrogen peroxide

5. (a) (i) What is rusting? ……
……..

(ii) State two necessary conditions for an iron nail to rust.
……. and ……..

(b) Briefly describe the following methods of preventing iron from rusting:

(i) Electroplating …….
…….

(ii) Galvanizing ……
…….

(iii) Sacrificial protection……
…….

(iv) Use of Silica gel ……
……..…

6. (a) Define 'solution' and give two examples of solution ;

Solution ……
…….
…….

(b) Give four differences a mixture and a compound

Mixture	Compound
(i)	
(ii)	
(iii)	
(iv)	

7. (a) Chemistry laboratory can be a dangerous place if safety measures are not adhered to. Assume you are given a chemistry laboratory to take care of, briefly explain any four of those safety measures you would take to make the laboratory safe to use.

(i) …… …….

(ii) …….

(iii) …… …….

(iv) …… …….

(b) A child of nine years old was having a nose bleed and you wanted to help. Outline three steps you would follow to administer a first aid to the child.

(i) ……..……..
(ii)…….……
(iii)…… ……

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8. (a) Give four differences between boiling and evaporation.

Boiling	Evaporation
(i)	
(ii)	
(iii)	
(iv)	

(b) Briefly explain the behavior of the particles of matter when in a solid state and when in a liquid state.

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……
……. …….
……

9. (a) Define each of the following:
   (i) Fire fighting …….. ……..
   (ii) Class A fires ……
   (iii) Class B fires ……
   (iv) Class C fires …….

(b) (i) Briefly explain why class B fires should not be extinguished by using water

……..
……..…….

(ii) Before using fire extinguishers to fight fires, there are some safety precautions you have to keep in mind. State any four of those safety precautions.

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SECTION C (15 MARKS)

10. Giving at least six points, explain the Importance of changing one state of matter to another.